



{ یک یادگیری کارساز }

بوتکمپ

Artificial Intelligence

MLOps

Session 8

Session Overview

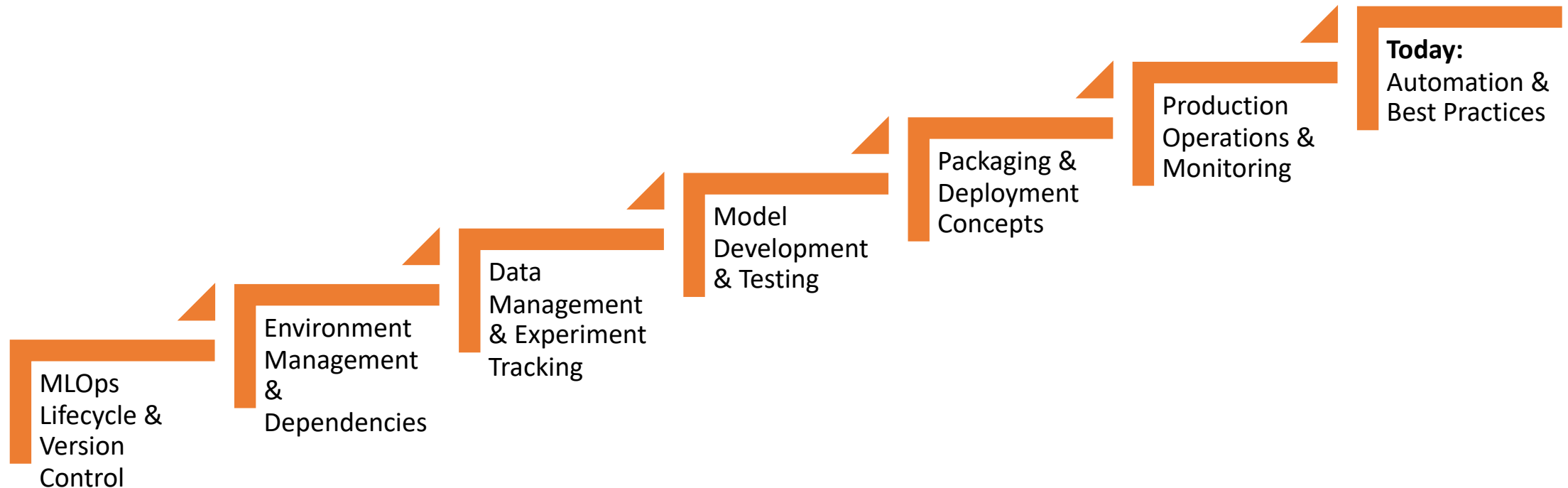
CI/CD & MLOps Best Practices

Today's Journey:

- Implement CI/CD pipelines for ML projects
- Establish automated testing and validation
- Build effective team collaboration workflows
- Explore MLOps career paths and next steps

Course Journey Recap

What We've Learned:



Continuous Integration for ML Projects

Traditional CI/CD:

- Code integration and testing
- Automated deployment
- Fast feedback loops

ML CI/CD Extends This:

- Code + Data + Model integration
- Multi-artifact deployment
- Performance-based validation
- Longer feedback cycles

VisionaryAI CI/CD Example

Computer Vision Team - Defect Detection:

- Code changes must not break functionality
- New training data maintains quality standards
- Updated models meet performance thresholds
- Deployment doesn't disrupt production monitoring

ML-Specific CI/CD Challenges

Data Dependencies:

- Changing data quality over time
- Seasonal variations and drift
- External data source reliability

Model Versioning:

- Code + weights + configuration
- Environment-specific performance
- Complex dependency chains

Testing Complexity:

- How to unit test neural networks?
- Long-running training processes
- Performance vs. accuracy trade-offs

ML CI Pipeline Components

Five Key Stages:

1. Code Quality Checks

- Linting, formatting, security

2. Unit Testing

- Functions, mocks, preprocessing

3. Integration Testing

- End-to-end pipeline validation

4. Model Validation

- Accuracy, bias, performance checks

5. Data Quality Checks

- Schema, drift, completeness

Code Quality Example - NLP Helpdesk

VisionaryAI Helpdesk Agent Pipeline

- Consistent code standards for customer data security
- Static type checking for reliability
- Documentation requirements for team collaboration
- Dependency scanning for vulnerabilities

Automated Testing Strategies

Property-Based Testing:

- Model predictions should be consistent
- Confidence scores between 0 and 1
- No null values after preprocessing

Metamorphic Testing:

- Image rotation → same defect classification
- Synonyms → similar query classification
- Irrelevant products → stable recommendations

Shadow Testing:

- Run new models alongside production
- Compare without affecting users
- Build deployment confidence

Deployment Automation & Team Collaboration

Deployment Strategies Overview

Four Main Approaches

Blue-Green: Two identical environments, instant switch

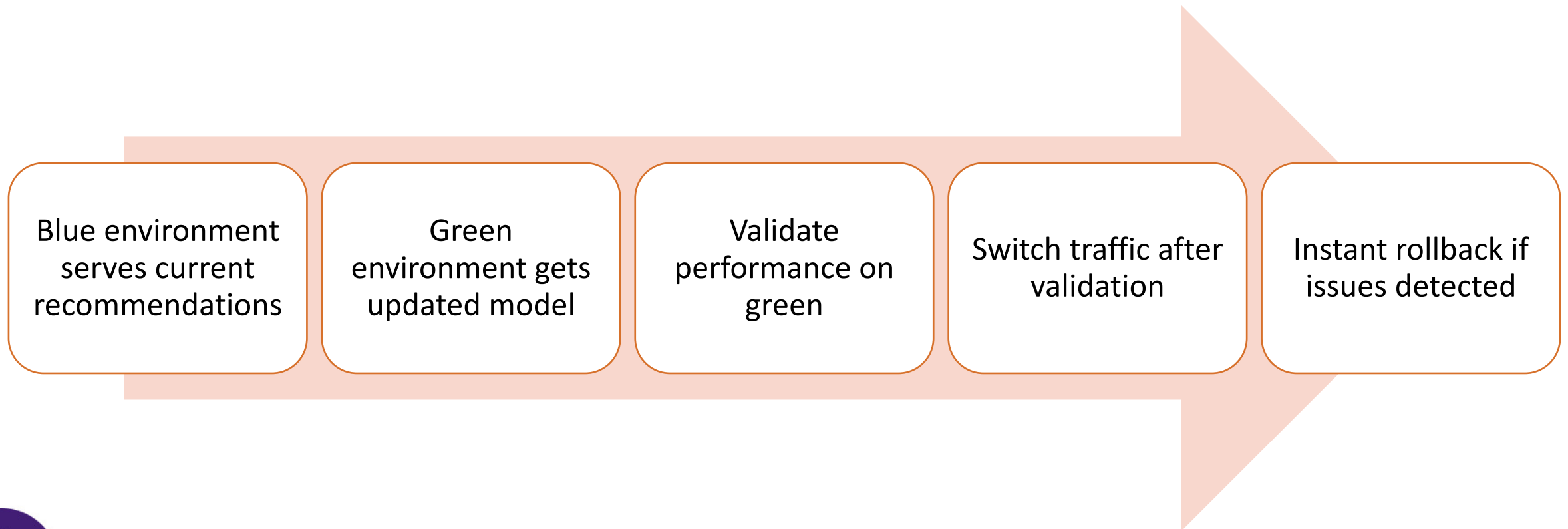
Canary: Gradual rollout to user subset

A/B Testing: Compare multiple versions

Rolling: Replace instances gradually

Blue-Green Example Recommender System

VisionaryAI Recommendation Engine



Deployment Automation Components

Infrastructure as Code:

- Version controlled infrastructure
- Reproducible deployments
- Automatic scaling



Model Registry Integration:

- Automated promotion through stages
- Approval processes for production
- Deployment history tracking



Health Checks & Monitoring:

- Deployment validation
- Performance monitoring
- Automatic rollback triggers

Team Collaboration Workflows

GitOps for ML Teams:

- Feature branch strategy
- Automated testing triggers
- Code review processes
- Clean merge history

VisionaryAI Example:

1. Feature branch: `feature/improved-cnn-architecture`
2. Local development and testing
3. CI pipeline triggered on push
4. Team code review
5. Merge after approval

Code Review Best Practices for ML

Review Checklist:

- ✓ Random seeds set for reproducibility?
- ✓ Appropriate train/validation/test split?
- ✓ No potential data leakage?
- ✓ Model complexity matches dataset size?
- ✓ Suitable evaluation metrics?

Cross-Functional Collaboration:

- Data Scientists ↔ ML Engineers
- DevOps Integration
- Shared monitoring systems

Documentation Strategy

Three Documentation Types

Model Documentation:

Model cards with performance metrics
Training data descriptions
Known limitations and biases

Process Documentation:

Deployment procedures
Incident response playbooks
Architecture decision records

Code Documentation:

Comprehensive docstrings
Setup instructions
API documentation

Knowledge Sharing Practices

Meeting Cadence:

- Weekly stand-ups
- Monthly technical deep-dives
- Quarterly retrospectives

Internal Tech Talks:

- Successful experiments
- New tools and techniques
- Production incident case studies

Documentation Culture:

- "Document as you go"
- Peer review for docs
- Regular audits
- Recognition for good documentation

Career Paths Section

Entry Level (0-2 years):

- Junior MLOps Engineer
- Data Engineer (ML-focused)
- Software Engineer (ML Platform)

Senior Level (5+ years):

- Senior MLOps Engineer
- ML Infrastructure Architect
- Head of ML Platform

Mid Level (2-5 years):

- MLOps Engineer
- ML Platform Engineer
- DevOps Engineer (ML-specialized)

Specialized Career Paths

Emerging Specializations:

ML Security Engineer

Security focus

ML Compliance Engineer

Regulatory compliance

MLOps Consultant

Help organizations adopt MLOps

ML Platform Product Manager

Product strategy for ML tools

Skills Development Roadmap

Foundation (Months 1-6):

- Python & software engineering
- Git version control
- Linux command line
- Basic cloud platforms
- Docker containers

Intermediate (Months 6-18):

- CI/CD tools
- Infrastructure as Code
- Kubernetes basics
- Monitoring tools
- ML frameworks

Advanced (18+ months):

- Kubernetes at scale
- Advanced cloud services
- Security & compliance
- Performance optimization

Building Your MLOps Portfolio

Project Ideas:

1. End-to-end ML pipeline
2. Model monitoring dashboard
3. CI/CD pipeline implementation
4. Infrastructure as Code deployment

Contribution Strategies:

- Open-source contributions
- Technical blog posts
- Conference presentations
- Reusable component sharing

Networking:

- MLOps communities
- Local meetups
- Hackathons
- Vendor communities

Code Demo

- CI/CD for ML

Course Completion & Next Steps

What You've Accomplished:

- ✓ MLOps fundamentals mastery
- ✓ End-to-end pipeline understanding
- ✓ Production-ready practices
- ✓ Career roadmap clarity

Your Next Actions:

- Apply concepts in personal projects
- Join MLOps communities
- Explore advanced topics
- Start building your portfolio

MLOps Fundamentals



Final Takeaways

Key Principles to Remember:

- Start small, automate incrementally
- Prioritize reliability over complexity
- Maintain a learning mindset
- Collaborate effectively with teams
- Document everything

The MLOps Journey Continues:

- Field is rapidly evolving
- New tools and practices emerging
- Community support is essential
- Practice makes permanent

Course Complete!

Thank you for joining [MLOps Fundamentals](#)

Continued Learning Resources:

- Advanced MLOps course (coming soon)
- Open-source projects
- The MLOps Community
- Designing Machine Learning Systems book
by Chip Huyen

Stay Connected:

- Course discussion forum
- GitHub repository
- LinkedIn





دانستگار

THANK YOU

FOR YOUR ATTENTION!